

WEB PROGRAMMING LAB-11 HOME EXERCISES

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Name: Devadathan N R

Registration No: 230905010

Class: CSE - A1

Roll No: 04

Title: Databases-Part II

Question 1

Create a web page with DropDownList, Textboxes and Buttons. Assume the table 'Human' with First name, Last name, Phone, Address and City as fields.

When the page is loaded, only first names will be displayed in the drop-down list. On selecting the name, other details will be displayed in the respective TextBoxes. On clicking the update button, the table will be updated with new entries made in the text box. On clicking the delete button, the selected record will be deleted from the table, and the DropDownList is refreshed.

Create project & App

```
django-admin startproject HumanProject
cd HumanProject
python manage.py startapp humanapp
```

HumanProject/settings.py

```
import os

INSTALLED_APPS = [
    'humanapp',
    'django.contrib.admin',
    'django.contrib.auth',
    'django.contrib.contenttypes',
    'django.contrib.sessions',
    'django.contrib.messages',
    'django.contrib.staticfiles',
]

STATIC_URL = '/static/'
STATICFILES_DIRS = [BASE_DIR / 'static']
```

Project URLs : HumanProject/urls.py

```
from django.contrib import admin
from django.urls import path, include
```

```
urlpatterns = [
    path('admin/', admin.site.urls),
    path('', include('humanapp.urls')),
]
```

App URLs: `humanapp/urls.py`

```
from django.urls import path
from . import views

urlpatterns = [
    path('', views.human_manage, name='human_manage'),
]
```

Models: `humanapp/models.py`

```
from django.db import models

class Human(models.Model):
    first_name = models.CharField(max_length=100)
    last_name = models.CharField(max_length=100)
    phone = models.CharField(max_length=15)
    address = models.CharField(max_length=255)
    city = models.CharField(max_length=100)

    def __str__(self):
        return self.first_name
```

Forms: `humanapp/forms.py`

```
from django import forms
from .models import Human

class HumanForm(forms.ModelForm):
    class Meta:
        model = Human
        fields = ['first_name', 'last_name', 'phone', 'address', 'city']
```

View: `humanapp/views.py`

```
from django.shortcuts import render, redirect, get_object_or_404
from .models import Human
from .forms import HumanForm

def human_manage(request):
    humans = Human.objects.all()
    selected_human = None
    form = HumanForm()

    # 1. Handle selection from Dropdown (GET request)
    human_id = request.GET.get('human_id')
```

```

if human_id and human_id.isdigit():
    selected_human = get_object_or_404(Human, id=human_id)
    form = HumanForm(instance=selected_human)

# 2. Handle Update and Delete (POST request)
if request.method == 'POST':
    curr_id = request.POST.get('curr_id')

    if curr_id:
        instance = get_object_or_404(Human, id=curr_id)
        if 'update' in request.POST:
            form = HumanForm(request.POST, instance=instance)
            if form.is_valid():
                form.save()
                return redirect('human_manage')
        elif 'delete' in request.POST:
            instance.delete()
            return redirect('human_manage')
    else:
        # Optional: Allow adding a new record if none is selected
        form = HumanForm(request.POST)
        if form.is_valid():
            form.save()
            return redirect('human_manage')

return render(request, 'human_manage.html', {
    'humans': humans,
    'form': form,
    'selected_human': selected_human
})

```

Presentation: `humanapp/templates/human_manage.html`

```

<!DOCTYPE html>
<html>
<head>
    <title>Human Management</title>
</head>
<body>
    <h2>Human Database Management</h2>

    <form method="GET" action="">
        <label>Select First Name:</label>
        <select name="human_id" onchange="this.form.submit()">
            <option value="">-- Choose a Person --</option>
            {% for h in humans %}
                <option value="{{ h.id }}" {% if selected_human.id == h.id %}selected
{% endif %}>
                    {{ h.first_name }}
            </option>
            {% endfor %}
        </select>
    </form>

```

```

<hr>

<form method="POST" action="">
    {% csrf_token %}
    <input type="hidden" name="curr_id" value="{{ selected_human.id|default:'' }}">

    <div style="margin-bottom: 10px;">
        {{ form.as_p }}
    </div>

    <button type="submit" name="update">Update Record</button>
    <button type="submit" name="delete" onclick="return confirm('Are you sure?')">
Delete Record</button>
    <a href="{% url 'human_manage' %}">Clear Form</a>
</form>

</body>
</html>

```

Execution

```

python manage.py makemigrations
python manage.py migrate
python manage.py runserver 8080

```

Output

New Entry

Human Database Management

Select First Name:

First name:

Last name:

Phone:

Address:

City:

[Clear Form](#)

User first name selected from dropdown. Details autofilled.

Human Database Management

Select First Name: 

First name:

Last name:

Phone:

Address:

City:

[Clear Form](#)

Old user selected and deleted

Human Database Management

Select First Name: 

First name:

Last name:

Phone:

Address:

City:

[Clear Form](#)

Human Database Management

Select First Name: -- Choose a Person -- ▾

First name: Narendra

Last name:

Phone:

Address:

City:

[Clear Form](#)

Question 2

Create a web page that receives the following information from a set of students: Student Id, Student Name, Course Name and Date of Birth. The application should also display the information of all the students once the data is Entered.

Create project & App

```
django-admin startproject StudentProject
cd StudentProject
python manage.py startapp studentapp
```

StudentProject/settings.py

```
import os

INSTALLED_APPS = [
    'studentapp',
    'django.contrib.admin',
    'django.contrib.auth',
    'django.contrib.contenttypes',
    'django.contrib.sessions',
    'django.contrib.messages',
    'django.contrib.staticfiles',
]
```

```
STATIC_URL = '/static/'
STATICFILES_DIRS = [BASE_DIR / 'static']
```

Project URLs : `StudentProject/urls.py`

```
from django.contrib import admin
from django.urls import path, include

urlpatterns = [
    path('admin/', admin.site.urls),
    path('', include('studentapp.urls')),
]
```

App URLs: `studentapp/urls.py`

```
from django.urls import path
from . import views

urlpatterns = [
    path('', views.student_entry, name='student_entry'),
]
```

Models: `studentapp/models.py`

```
from django.db import models

class Student(models.Model):
    student_id = models.CharField(max_length=20, primary_key=True)
    student_name = models.CharField(max_length=100)
    course_name = models.CharField(max_length=100)
    dob = models.DateField() # Date of Birth field [cite: 400]

    def __str__(self):
        return self.student_name
```

Forms: `studentapp/forms.py`

```
from django import forms
from .models import Student

class StudentForm(forms.ModelForm):
    class Meta:
        model = Student
        fields = ['student_id', 'student_name', 'course_name', 'dob']
        # Adding a widget to make the DOB field a date picker
        widgets = {
            'dob': forms.DateInput(attrs={'type': 'date'}),
        }
```

View: `studentapp/views.py`

```

from django.shortcuts import render, redirect
from .models import Student
from .forms import StudentForm

def student_entry(request):
    # Retrieve all student records for display [cite: 469]
    students = Student.objects.all()
    form = StudentForm()

    if request.method == 'POST':
        form = StudentForm(request.POST)
        if form.is_valid():
            form.save() # Save the data into the db [cite: 463, 539]
            return redirect('student_entry')

    return render(request, 'student_page.html', {
        'form': form,
        'students': students
    })

```

Presentation: `studentapp/templates/student_page.html`

```

<!DOCTYPE html>
<html>
<head>
    <title>Student Information System</title>
</head>
<body>
    <h2>Enter Student Details</h2>
    <form method="POST">
        {% csrf_token %}
        {{ form.as_p }}
        <button type="submit">Submit Data</button>
    </form>

    <hr>

    <h2>Registered Students List</h2>
    <table border="1">
        <thead>
            <tr>
                <th>ID</th>
                <th>Name</th>
                <th>Course</th>
                <th>Date of Birth</th>
            </tr>
        </thead>
        <tbody>
            {% for student in students %}
            <tr>
                <td>{{ student.student_id }}</td>
                <td>{{ student.student_name }}</td>
                <td>{{ student.course_name }}</td>
            </tr>
            {% endfor %}
        </tbody>
    </table>

```



```

        <td>{{ student.dob }}</td>
    </tr>
    {% empty %}
    <tr>
        <td colspan="4">No student records found.</td>
    </tr>
    {% endfor %}
</tbody>
</table>
</body>
</html>

```

Execution

```

python manage.py makemigrations
python manage.py migrate
python manage.py runserver 8080

```

Output

Enter Student Details

Student id:

Student name:

Course name:

Dob: 

Registered Students List

ID	Name	Course	Date of Birth
1	Akshay	WPL	April 5, 2026
